Lab 02

Due Feb 3 5:00 PM PST

STAT240 Spring 2025

2026-01-30

Instructions

For this homework, provide a rendered R Markdown file in PDF format. You may render the R Markdown file to HTML and then convert the HTML file to PDF using the print function in a web browser. Indicate your student number on the markdown file and make a section for each problem. There is a qmd file for you to use on canvas.

Problem 1: My Median

a) Write an R function called `my_median`. This function should take a single parameter (a vector) and return the median of that vector. Write this function without using the built-in R command `median`. Provide the code for this function.

(4 points)

```
# Example R code
define_my_median <- function(vector) {
  # Code logic here
}
```

b) Consider the stock prices in the file `AS-N100.tsv` provided in the archive for week 3. These data involve stock prices for 27 symbols (in the column `ticker`). Compute your student number modulo 27 and add one, call this value `a` (i.e., `a` is a number between 1 and 27 inclusive). Imagine a list of the 27 symbols in alphabetical order. You will examine the `a`-th symbol in this list. You may find the symbol using the R code `sort(unique(data$ticker))[a]`. Compute the median opening price of the symbol you examine over all rows in `AS-N100.tsv` involving that symbol, using your `my_median` function. Provide all code and output.

(2 points)

```r
# Example R code to read and process
library(readr)
data <- read_tsv("AS-N100.tsv")
symbols <- sort(unique(data$ticker))
# Compute value of 'a'
# Example of custom median application
```

Problem 2: X Marks the Spot

a) The image `Figure03.png` (provided in the canvas folder for this lab) is a satellite photo of Vancouver and the surrounding area. In this problem, you will mark the location of a library on this image automatically using code.

The upper left-hand corner of this image has approximate GPS coordinates `49.410705,-124.217671`. The lower right-hand corner of this image has approximate GPS coordinates `47.929083,-121.994887`. Consider the file `libraries.json` provided in the archive for week 3. This file contains information about 21 branches of the Vancouver Public Library.

Write R code to display `Figure03.png` and to mark an `X` symbol on the GPS coordinates of the `b`-th library branch. Print the name and address of the library branch directly below the `X`. Here, `b` is your student number modulo 21 plus one (i.e., `b` is a number between 1 and 21 inclusive).

Note that after loading the library `rjson` and loading `libraries.json` using:

```r
libraries <- fromJSON(file = "libraries.json")
```

the coordinates of the `b`-th branch can be accessed with:

```r
libraries[["features"]][[b]][["geometry"]]$coordinates
```

The branch name and address can be accessed with:

```r
libraries[["features"]][[b]][["properties"]]$maptip
```

Provide your code and the resulting display. **Note that you may find it is quite slow to load and display the image when you test it in the script, and also buggy. It should run correctly when you render it, but it will also be slow.**

(4 points)